Mehedi Hasan Nirzan

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Research Interests

Cardiovascular Biomechanics, Patient-Specific CFD, Cardiovascular Digital Twins, and Physics-Informed Machine Learning.

Education

Bangladesh University of Engineering and Technology (BUET) Dhaka, Bangladesh
Bachelor of Science in Biomedical Engineering CGPA: 3.59/4.00

Research and Projects

Prediction of Left Coronary Artery Plaque Rupture Risk using CCTA-based One-Way FSI
Undergraduate Thesis | Supervisor: Dr. Muhammad Tarik Arafat, Professor, BME, BUET *2025 – Present*

- Developed a patient-specific FSI framework integrating ANSYS Fluent CFD and Transient Structural analysis on CCTA-derived arterial geometries reconstructed using Materialise Mimics and 3-matic.
- Modeled multi-layer plaque components (fibrous cap, necrotic core, calcification) as geometrically and materially distinct bodies using hyperelastic Yeoh material properties to assess mechanical vulnerability and stress distribution.

Impact of Image Processing on Retinal Lesion Classification from OCT B-Scans 2025
Deep Learning Project | PI: Samiul Based Shuvo, Assistant Professor, BME, BUET Presentation

- Designed a CLAHE-based preprocessing pipeline achieving 97% accuracy and a 4.75% F1-score improvement using VGG16 across multiple datasets.

Fingerprint Recognition System for Securing Patient Health Data 2025
Signal Processing Project | PI: Dr. Md. Kamrul Hasan, Professor, EEE, BUET Presentation

- Developed a DSP-based biometric authentication system in MATLAB to secure sensitive patient data, including ECG signals.

Mpox Skin Lesion Detection System
Deep Learning Project | PI: Shams Nafisa Ali, Assistant Professor, BME, BUET *2024*

- Built a Python-based deep learning web application for automated Mpox lesion detection, improving diagnostic accuracy across diverse populations.

Cost-Effective Video Laryngoscope for Resource-Limited Clinical Settings 2024
Capstone Project | PI: Abdullah Al Mamun, Lecturer, BME, BUET Presentation

- Designed and 3D-printed a PLA-based video laryngoscope in built-in-display and smartphone-integrated configurations at an estimated prototype cost of $80.

Fractional-Order Control for Anesthesia and Hemodynamic Stabilization 2024
Biomedical Control Systems Project | PI: Shams Nafisa Ali, Assistant Professor, BME, BUET Presentation

- Implemented a decentralized–decoupled fractional-order control system in MATLAB/Simulink for anesthesia and hemodynamic stabilization.
- Compared simulated BIS, MAP, cardiac output, and drug-infusion responses with published results.

Helmet Positional Stability Testing Setup (Roll-Off Test)
2023

- Designed and fabricated an aluminium test rig in SolidWorks for standardized helmet safety and positional stability evaluation.

Experience

Bioinnovation Lab — Dhaka, Bangladesh
Research Assistant — *July 2026 – Present*

- Mentoring two undergraduate students in their thesis research.

Cortex Advisory — Dhaka, Bangladesh
Research Engineer — *August 2026 – Present*

- Contributed to the preparation of a research project proposal, including technical planning and proposal development.

Square Hospitals Ltd. — Panthapath, Dhaka, Bangladesh
Biomedical Engineering Intern — *9 March 2025 – 24 March 2025*

- Gained practical exposure to biomedical equipment management, clinical engineering practices, and hospital-based medical device operations.

Achievements

- Finalist, Rice University RICE360 Global Health Technologies Design Competition – Houston, Texas — 2025
- 1st Place, Project Showcasing Competition, 1st National Biomedical Engineering Conference, JUST — 2025
- Selected Presenter, Carnegie Mellon Forum on Biomedical Engineering — 2025
- 1st Prize, BME 202 Biomechanics Design Contest (Ranked 1st among 5 groups) — 2022

Technical Skills

Programming and Technical Computing: Python, MATLAB, C/C++, LaTeX
CAD and Modeling: SolidWorks, Materialise Mimics, Materialise 3-matic, Proteus
Simulation: ANSYS, COMSOL Multiphysics, Simulink
Hardware and Embedded Systems: Arduino, Microcontrollers, Raspberry Pi
Graphing Software: OriginPro

Leadership and Involvement

- Help BUETian | General Secretary — July 2025 – June 2026
- Biomedical Engineering Student Chapter (BMES) | Secretary, Outreach — May 2025 – May 2026
- Ankur International | Project Director — May 2024 – May 2025

Relevant Coursework

Biomechanics, Biofluid Mechanics and Heat Transfer, Biomedical Transport Fundamentals, Numerical Techniques, Biomedical Control Systems, Medical Imaging, Machine Learning for Biomedical Engineers, Digital Signal Processing, Probability and Statistics, Linear Algebra, Differential Equations